

# LOW-TENSION CABLE DYNAMICS: NUMERICAL AND EXPERIMENTAL STUDIES

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**ABSTRACT:** An efficient and robust numerical method is presented for the dynamic analysis of low-tension cables. The numerical solution strategy is based on finite-difference approximations of differential equations. In a scheme used by other researchers, known as the box scheme, the trapezoidal method is employed in both space and time domains. This scheme, however, gives rise to spurious high-frequency oscillations in cable tension response, as discovered in the research work reported herein. A modified box scheme is proposed to eliminate the problem. To improve computational efficiency, an iterative procedure is used to solve the resulting nonlinear simultaneous equations. A "free-fall" problem of cable dynamics involving low tension and large displacement motion is studied numerically. An experimental program is carried out to verify the accuracy of the numerical solution with regards to cable tension response.

## INTRODUCTION

Cables may range from several kilometers long in the case of cable suspension bridges to less than a meter long in the case of musical string bridges. Capable of transmitting forces, carrying payload, and conducting electrical power and signals across large distances, cables are normally held in position by tensions along their length and end forces at supports. In most structural engineering applications, cables experience high tension, such as in cable-stayed and cable suspension structures. Typically, these taut cables have sag-to-span ratios of less than 1:8 (Perkins and Mote 1987). The equilibrium shape of small sag cables can be approximated by a parabola that greatly simplifies the analysis. Linear theory can be applied due to the relatively small displacement motion, and analytical solutions may even be available in terms of natural frequencies and mode shapes. Extensive research has been carried out on the analysis of high-tension cables for several decades [see, e.g., Simpson (1972), Blik and Triantafyllou (1985), Perkins and Mote (1987), Rao and Iyengar (1991), and Khala (1995)].

Low-tension cables, on the other hand, have received relatively less attention. Examples of low-tension cables include power and signal transmission cables and underwater cables and mooring lines (due to buoyancy effect) in offshore applications. From the computational point of view, they pose a greater challenge than high-tension cables, particularly when dynamic analysis is involved. It is often necessary to use nonlinear analysis to deal with the generally large displacement motion of low-tension cables. Another important aspect is that the flexural stiffness has to be considered, without which geometric stiffness is lost when the tension approaches zero, leading to a numerical problem in convergence.

One of the earliest research works on low-tension cable dynamics was by Leonard and Recker (1972) who presented a method of solution for the three-dimensional nonlinear deployment problem of a single, lightly stressed cable. Axial and bending deformations were neglected. Based on the theory of incremental deformation, the governing equations were re-

duced to a system of quasi-linear equations dependent on the prior history of the deployment. Blik and Triantafyllou (1985) found that the geometric nonlinearity due to large amplitude motions could cause a substantially large nonlinear dynamic tension oscillating at twice the frequency of excitation. Similar results were reported in an experimental study by Welch and Tulin (1993).

Ablow and Schechter (1983) used a finite-difference (FD) approximation, called the box scheme, which essentially applied the trapezoidal method (Gear 1971) in both time and space domains, to simulate the motion of a towed cable. Numerical results for displacements were found to agree well with the experimental results, but tension response was not reported.

More recent works include dynamic analysis of hanging strings (Triantafyllou and Triantafyllou 1991) and a few other related papers (Triantafyllou and Howell 1992, 1993a,b). These works have shown that the equations of motion used for solving high-tension problems may become singular for low-tension cable problems when the cable tension vanishes anywhere along the cable. By including selected terms containing higher-order spatial derivatives, they found that the problem associated with the loss of tension could be alleviated. One obvious way is to include the flexural stiffness in the governing equations (Triantafyllou and Howell 1993b). For the same reason, Burgess (1992) derived a set of equations of motion for submerged cable accounting for bending and torsional stiffness terms. Most of these studies, however, ignored axial elongation of the cable (i.e., inextensible). This assumption may not be appropriate for some practical problems in which cable stretching has a pronounced effect on its response, particularly if the cable has low modulus of elasticity (e.g., elastomeric material).

## FD and Finite-Element (FE) Methods

In general, the governing equations for low-tension cable dynamics are coupled and highly nonlinear. Analytical solutions are unavailable and numerical means have to be used. The FD method and FE method are two widely used numerical methods to solve such nonlinear dynamics problems. The FD method approximates the governing differential equations by some difference equations, whereas the FE method discretizes the continuum (cable) into a finite number of elements. In practice, each method has its merits and limitations. The main advantage of the FE method is that the procedure can be implemented in general purpose analysis programs for a wide range of applications. In contrast, the FD method is more suitable for some specific situations where numerical instability may pose convergence problem to the FE method. In this paper, the FD method is adopted to study the large displacement

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motion of a single span cable with low tensions in a two-dimensional vertical plane.

## FORMULATION

The following assumptions are made in the formulation:

1. Plane sections of cable remain plane after deformation.
2. The cable material is homogeneous, isotropic, and linearly elastic.
3. Shear deformation and Poisson's effect are neglected.
4. Cable strain is small though the displacement and rotation may be large.

To improve numerical stability as mentioned earlier, the effect of flexural stiffness is included in the formulation though the bending moment may be small. The slenderness ratio of the cable as given by  $\lambda = L/\sqrt{I/A}$  is usually large, where  $L$  = length of the cable;  $A$  = cross-sectional area; and  $I$  = moment of inertia of the cross section. The axial strain is given by

$$e = \frac{ds' - ds}{ds} \quad (1)$$

where  $ds$  and  $ds'$  = original and deformed lengths, respectively, of a differential element as shown in Fig. 1. The following equations of motion can be derived:

$$\frac{\partial f_x}{\partial s} = \rho A \frac{\partial v_x}{\partial t} - w_x(1 + e) + cv_x \quad (2)$$

$$\frac{\partial f_y}{\partial s} = \rho A \frac{\partial v_y}{\partial t} - w_y(1 + e) + \rho Ag + cv_y \quad (3)$$

$$\frac{\partial m}{\partial s} = (1 + e)(f_x \sin \theta - f_y \cos \theta) \quad (4)$$

where  $t$  = time;  $m$  = bending moment;  $\theta$  = rotation;  $c$  = damping constant to account for energy loss;  $f$  = cable force;  $w$  = applied distributed force (per unit deformed length); and  $v$  = velocity. Subscripts  $x$  and  $y$  refer to the horizontal and vertical components, respectively. Rotational inertia effect is neglected in (4). Considering both elongation and bending effects of the cable, the constitutive relations are

$$e = \frac{1}{EA} (f_x \cos \theta + f_y \sin \theta) \quad (5)$$

$$\frac{\partial \theta}{\partial s} = \frac{m}{EI} \quad (6)$$

The nonlinear kinematic relations are (Fig. 1)

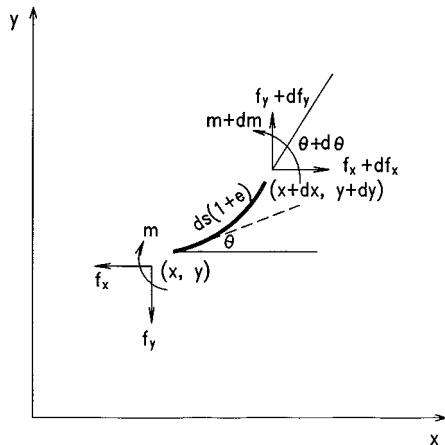


FIG. 1. Geometry and Internal Forces of Differential Element

$$\frac{\partial x}{\partial s} = (1 + e) \cos \theta \quad (7)$$

$$\frac{\partial y}{\partial s} = (1 + e) \sin \theta \quad (8)$$

Differentiating (7) and (8) with respect to time and using (5) leads to

$$\begin{aligned} \frac{\partial v_x}{\partial s} &= \frac{\cos^2 \theta}{EA} \frac{\partial f_x}{\partial t} + \frac{\cos \theta \sin \theta}{EA} \frac{\partial f_y}{\partial t} \\ &+ \left( -\frac{f_x \sin 2\theta}{EA} + \frac{f_y \cos 2\theta}{EA} - \sin \theta \right) \frac{\partial \theta}{\partial t} \end{aligned} \quad (9)$$

$$\begin{aligned} \frac{\partial v_y}{\partial s} &= \frac{\sin \theta \cos \theta}{EA} \frac{\partial f_x}{\partial t} + \frac{\sin^2 \theta}{EA} \frac{\partial f_y}{\partial t} \\ &+ \left( \frac{f_x \cos 2\theta}{EA} + \frac{f_y \sin 2\theta}{EA} + \cos \theta \right) \frac{\partial \theta}{\partial t} \end{aligned} \quad (10)$$

The following dimensionless variables are introduced for convenience:

$$\bar{x} = \frac{x}{L}; \quad \bar{y} = \frac{y}{L}; \quad \bar{s} = \frac{s}{L}; \quad \bar{v}_x = \frac{v_x}{\sqrt{gL}}; \quad \bar{v}_y = \frac{v_y}{\sqrt{gL}}; \quad \bar{f}_x = \frac{f_x}{EA};$$

$$\bar{f}_y = \frac{f_y}{EA}; \quad \bar{m} = \frac{m}{EAL}; \quad \bar{\mu} = \frac{\rho g L}{E}; \quad \bar{w}_x = \frac{w_x L}{EA};$$

$$\bar{w}_y = \frac{w_y L}{EA}; \quad \bar{c} = \frac{c L \sqrt{gL}}{EA}; \quad \bar{t} = \frac{t}{\sqrt{L/g}}$$

Eqs. (2)–(4), (6), (9), and (10) can be expressed in matrix form as follows:

$$\frac{\partial \mathbf{u}}{\partial s} = \mathbf{A} \frac{\partial \mathbf{u}}{\partial t} + \mathbf{B} \quad (11)$$

where

$$\mathbf{u} = \{\bar{v}_x \quad \bar{v}_y \quad \bar{f}_x \quad \bar{f}_y \quad \bar{m}\}^T \quad (12)$$

is the (normalized) response vector, and

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & (-\bar{f}_x \sin 2\theta + \bar{f}_y \cos 2\theta - \sin \theta) & \cos^2 \theta & \sin \theta \cos \theta & 0 \\ 0 & 0 & (\bar{f}_x \cos 2\theta + \bar{f}_y \sin 2\theta + \cos \theta) & \sin \theta \cos \theta & \sin^2 \theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \bar{\mu} & 0 & 0 & 0 & 0 & 0 \\ 0 & \bar{\mu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (13)$$

$$\mathbf{B} = \left\{ \begin{array}{c} 0 \\ 0 \\ \bar{m} \\ \bar{w}_x(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \\ \bar{\mu} + \bar{c} \bar{v}_y - \bar{w}_y(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \\ \lambda^2(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta)(\bar{f}_x \sin \theta - \bar{f}_y \cos \theta) \end{array} \right\} \quad (14)$$

Because matrices  $\mathbf{A}$  and  $\mathbf{B}$  are functions of the response vector  $\mathbf{u}$ , the preceding system of equations is nonlinear and thus has to be solved by numerical means.

## NUMERICAL SOLUTION TECHNIQUES

### Box Scheme

First, an FD scheme known as the box scheme (Keller 1970; Ablow and Schechter 1983) is adopted. The partial derivatives are approximated by using the trapezoidal method (Gear 1971)

in both time and spatial domains. Let  $\Delta \bar{t}$  and  $\Delta \bar{s}$  be the time step and the spatial step, respectively. Subscripts  $i$  and  $j$  correspond to discrete time point  $\bar{t}_i = i\Delta \bar{t}$  and space node  $\bar{s}_i = i\Delta \bar{s}$ , respectively. Applying the trapezoidal method at the midpoint of the time interval, we obtain

$$\left(\frac{\partial \mathbf{u}}{\partial s}\right)_{i+1} + \left(\frac{\partial \mathbf{u}}{\partial s}\right)_i = (\mathbf{A}_{i+1} + \mathbf{A}_i) \frac{\mathbf{u}_{i+1} - \mathbf{u}_i}{\Delta \bar{t}} + (\mathbf{B}_{i+1} + \mathbf{B}_i) \quad (15)$$

Similarly, applying the trapezoidal method at the midpoint between two space nodes gives

$$\frac{\mathbf{u}_{i+1,j+1} - \mathbf{u}_{i+1,j}}{\Delta \bar{s}} + \frac{\mathbf{u}_{i,j+1} - \mathbf{u}_{i,j}}{\Delta \bar{s}} = \frac{1}{2} (\mathbf{C}_{i,j+1} + \mathbf{C}_{i,j}) \quad (16)$$

where

$$\mathbf{C}_{i,j} = (\mathbf{A}_{i+1,j} + \mathbf{A}_{i,j}) \frac{\mathbf{u}_{i+1,j} - \mathbf{u}_{i,j}}{\Delta \bar{t}} + (\mathbf{B}_{i+1,j} + \mathbf{B}_{i,j}) \quad (17)$$

Forming the basis of the box scheme, (16) can also be written as

$$\mathbf{P}_{i,j} \mathbf{u}_{i+1,j+1} + \mathbf{Q}_{i,j} \mathbf{u}_{i+1,j} = \mathbf{R}_{i,j} \quad (18)$$

where

$$\mathbf{P}_{i,j} = \mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} (\mathbf{A}_{i+1,j+1} + \mathbf{A}_{i,j+1}) \quad (19)$$

$$\mathbf{Q}_{i,j} = -\mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} (\mathbf{A}_{i+1,j} + \mathbf{A}_{i,j}) \quad (20)$$

$$\begin{aligned} \mathbf{R}_{i,j} = & \left[ -\mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} (\mathbf{A}_{i+1,j+1} + \mathbf{A}_{i,j+1}) \right] \mathbf{u}_{i,j+1} \\ & + \left[ \mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} (\mathbf{A}_{i+1,j} + \mathbf{A}_{i,j}) \right] \mathbf{u}_{i,j} \\ & + \frac{\Delta \bar{s}}{2} (\mathbf{B}_{i+1,j+1} + \mathbf{B}_{i+1,j} + \mathbf{B}_{i,j+1} + \mathbf{B}_{i,j}) \end{aligned} \quad (21)$$

and  $\mathbf{I}$  = identity matrix of  $6 \times 6$ .

For cable dynamics problems, this scheme was first adopted perhaps by Ablow and Schechter (1983) and subsequently by some other researchers (Milinazzo et al. 1987; Burgess 1991;

$$\hat{\mathbf{B}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \bar{c} & 0 & 0 & 0 & 0 & 0 \\ 0 & \bar{c} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda^2(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \sin \theta & -\lambda^2(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \cos \theta & 0 \end{bmatrix} \quad (28)$$

Howell 1991; Triantafyllou and Howell 1992; Grosenbaugh et al. 1993). In these research works, the results were presented for cable configurations but not cable tension response. In the course of our study, we have found that the tension response obtained by using the box scheme contains a spurious high-frequency mode that is dependent on the time step used. This is due to the weak instability introduced by central difference approximation (Smith 1985) and will be illustrated in the numerical example to follow.

### Modified Box Scheme

To resolve this problem, an alternative scheme is proposed here by using backward difference approximation of  $\partial \mathbf{u} / \partial t$  in (11) in the time domain, leading to

$$\left(\frac{\partial \mathbf{u}}{\partial s}\right)_{i+1} = \mathbf{A}_{i+1} \frac{\mathbf{u}_{i+1} - \mathbf{u}_i}{\Delta \bar{t}} + \mathbf{B}_{i+1} \quad (22)$$

In the space domain that is bounded (unlike in the time domain) and constrained by boundary conditions, the trapezoidal method is still used. This gives rise to the same equation as (16), but with a different expression for  $\mathbf{C}_{i,j}$  as follows:

$$\mathbf{C}_{i,j} = \mathbf{A}_{i+1,j} \frac{\mathbf{u}_{i+1,j} - \mathbf{u}_{i,j}}{\Delta \bar{t}} + \mathbf{B}_{i+1,j} \quad (23)$$

Hence, (18) can be used for the modified box scheme, but with the following expressions:

$$\mathbf{P}_{i,j} = \mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} \mathbf{A}_{i+1,j+1} \quad (24)$$

$$\mathbf{Q}_{i,j} = -\mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} \mathbf{A}_{i+1,j} \quad (25)$$

$$\begin{aligned} \mathbf{R}_{i,j} = & -\frac{1}{2} \frac{\Delta \bar{s}}{\Delta \bar{t}} (\mathbf{A}_{i+1,j+1} \mathbf{u}_{i,j+1} + \mathbf{A}_{i+1,j} \mathbf{u}_{i,j}) \\ & + \frac{\Delta \bar{s}}{2} (\mathbf{B}_{i+1,j+1} + \mathbf{B}_{i+1,j}) \end{aligned} \quad (26)$$

### Iterative Method to Solve Nonlinear Equations

Eq. (18) represents a set of nonlinear equations to be solved, together with known boundary conditions, for  $\mathbf{u}$  at all nodes. A commonly adopted method is Newton-Raphson's method (Atkinson 1978), which is a linearization method. The advantage of Newton-Raphson's method is that its convergence rate is quadratic when near the true solution. This method was used to solve the nonlinear equations in the study of a towed cable system (Milinazzo et al. 1987) and other cable dynamics problems (Burgess 1991; Howell 1991). Newton-Raphson's method, however, requires the computation and storage of the Jacobian matrix at each time step. If the size of this matrix is large, the computation would be time consuming.

An alternative method is proposed next, to solve the system of nonlinear equations iteratively. To expedite numerical convergence, it is found useful to separate vector  $\mathbf{B}$ , which appears on the right-hand side of (18), into two parts as follows:

$$\mathbf{B} = \hat{\mathbf{B}} \mathbf{u} + \tilde{\mathbf{B}} \quad (27)$$

where

$$\tilde{\mathbf{B}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\bar{w}_x(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \\ \mu - \bar{w}_y(1 + \bar{f}_x \cos \theta + \bar{f}_y \sin \theta) \\ 0 \end{bmatrix} \quad (29)$$

It is advantageous to move the unknowns (i.e.,  $\hat{\mathbf{B}} \mathbf{u}$ ) to the left-hand side of (18) in the iterative procedure. For the box scheme, (27) is substituted into (21), and then into (18), leading to

$$\hat{\mathbf{P}}_{i,j} \mathbf{u}_{i+1,j+1} + \hat{\mathbf{Q}}_{i,j} \mathbf{u}_{i+1,j} = \hat{\mathbf{R}}_{i,j} \quad (30)$$

where

$$\hat{\mathbf{P}}_{i,j} = \mathbf{P}_{i,j} - \frac{\Delta \bar{s}}{2} \hat{\mathbf{B}}_{i+1,j+1} \quad (31)$$

$$\hat{\mathbf{Q}}_{i,j} = \mathbf{Q}_{i,j} - \frac{\Delta \bar{s}}{2} \hat{\mathbf{B}}_{i+1,j} \quad (32)$$

$$\begin{aligned} \hat{\mathbf{R}}_{i,j} = & \left[ -\mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta t} (\mathbf{A}_{i+1,j+1} + \mathbf{A}_{i,j+1}) \right] \mathbf{u}_{i,j+1} \\ & + \left[ \mathbf{I} - \frac{1}{2} \frac{\Delta \bar{s}}{\Delta t} (\mathbf{A}_{i+1,j} + \mathbf{A}_{i,j}) \right] \mathbf{u}_{i,j} \\ & + \frac{\Delta \bar{s}}{2} (\tilde{\mathbf{B}}_{i+1,j+1} + \tilde{\mathbf{B}}_{i+1,j} + \mathbf{B}_{i,j+1} + \mathbf{B}_{i,j}) \end{aligned} \quad (33)$$

where  $\mathbf{P}_{i,j}$  and  $\mathbf{Q}_{i,j}$  are given in (19) and (20), respectively. Discretizing the cable into  $n$  segments and applying (33) at  $j = 0, 1, 2, \dots, n-1$  gives the following matrix equation:

$$\begin{bmatrix} \hat{\mathbf{Q}}_{i,0} & \hat{\mathbf{P}}_{i,0} & \hat{\mathbf{P}}_{i,1} & \dots & 0 \\ & \hat{\mathbf{Q}}_{i,1} & \hat{\mathbf{P}}_{i,1} & \dots & 0 \\ & 0 & & \dots & \hat{\mathbf{Q}}_{i,n-1} & \hat{\mathbf{P}}_{i,n-1} \end{bmatrix} \begin{Bmatrix} \mathbf{u}_{i+1,0} \\ \mathbf{u}_{i+1,1} \\ \vdots \\ \mathbf{u}_{i+1,n} \end{Bmatrix} = \begin{Bmatrix} \hat{\mathbf{R}}_{i,0} \\ \hat{\mathbf{R}}_{i,1} \\ \vdots \\ \hat{\mathbf{R}}_{i,n-1} \end{Bmatrix} \quad (34)$$

For the modified box scheme, the same equations are obtained except for different expressions as follows:

$$\hat{\mathbf{P}}_{i,j} = \mathbf{P}_{i,j} - \frac{\Delta \bar{s}}{2} [\hat{\mathbf{B}}]_{i+1,j+1} \quad (35)$$

$$\hat{\mathbf{Q}}_{i,j} = \mathbf{Q}_{i,j} - \frac{\Delta \bar{s}}{2} [\hat{\mathbf{B}}]_{i+1,j} \quad (36)$$

$$\begin{aligned} \hat{\mathbf{R}}_{i,j} = & -\frac{1}{2} \frac{\Delta \bar{s}}{\Delta t} (\mathbf{A}_{i+1,j+1} \mathbf{u}_{i,j+1} + \mathbf{A}_{i+1,j} \mathbf{u}_{i,j}) \\ & + \frac{\Delta \bar{s}}{2} (\hat{\mathbf{B}}_{i+1,j+1} + \hat{\mathbf{B}}_{i+1,j}) \end{aligned} \quad (37)$$

where  $\mathbf{P}_{i,j}$  and  $\mathbf{Q}_{i,j}$  for the modified box scheme are given in (24) and (25), respectively.

Note that the matrix on the left-hand side of (34) is of size  $6n \times 6(n+1)$ . Eq. (34) thus represents a system of  $6n$  simultaneous nonlinear equations to be solved for six unknowns at each of the  $(n+1)$  nodes. Six additional equations are obtained by imposing known boundary conditions.

At each time step  $(i+1)$ , the results obtained in the previous time step,  $\mathbf{u}_{i,j}$ , are first used to compute  $\hat{\mathbf{Q}}_{i,j}$ ,  $\hat{\mathbf{P}}_{i,j}$ , and  $\hat{\mathbf{R}}_{i,j}$ , and (34) is solved by the Gauss elimination method to obtain a trial solution of  $\mathbf{u}_{i+1,j}$ . Eq. (34), though not symmetric, is narrowly banded with a maximum bandwidth of 12, which is exploited to expedite the equation solving. The trial solution is then used to update  $\hat{\mathbf{Q}}_{i,j}$ ,  $\hat{\mathbf{P}}_{i,j}$ , and  $\hat{\mathbf{R}}_{i,j}$ , and the equation solving is carried out again to improve the solution. The iterative procedure is repeated until convergence.

The iterative method avoids forming the Jacobian matrix. Generally speaking, the initial values should be sufficiently accurate for iterative schemes to work or converge rapidly. This is not a hindrance for cable dynamics, as accurate initial values are normally obtained a priori by static analysis to determine the static configuration of the cable.

## EXAMPLE PROBLEM: FREE FALL OF CABLE

A problem of cable free fall is studied that involves large displacement motion and low tension at the falling end. The results are obtained by applying the box scheme and the proposed modified box scheme, and they are compared with published results for cable shapes. To ascertain the accuracy of

tension response, a physical experiment of a rubber cable is conducted.

Consider a cable suspended between two pinned ends that is suddenly released at one end. The problem was first studied by Fried (1982) using quadratic cable elements and neglecting flexural stiffness. The effect of cable extension was considered in his paper, but the axial strain was assumed to be constant (0.01) throughout the cable length. This is an approximation since cable extension is proportional to tension force that is not constant. Damping is not considered in this case ( $c = 0$ ). The computed shapes of the falling cable were reportedly in good agreement with the experimental results captured by video camera, but the tension force was not presented by Fried (1982).

## Cable Shapes

The same problem is now analyzed here by using the box scheme and the modified box scheme. Computer programs are written to implement the box scheme and the modified box scheme, using the proposed iterative method to solve the resulting nonlinear algebraic equations. In the paper by Fried (1982), the horizontal span between the supports is  $D_L = 0.881$ , and the maximum sag ratio (sag over  $D_L$ ) is 0.207. Material properties of the cable and some other parameters are, however, not given. A static analysis is therefore carried out with the same horizontal span and, by trial and error, other parameters to obtain a sag of 0.2134 that is reasonably close to Fried's value. The parameters used here are  $EA = 1.57 \times 10^5$ ,  $\mu = 1.155 \times 10^{-2}$ , and  $\lambda = 679.7$ .

In the dynamic analysis, 41 space nodes (40 segments) along the cable length and time step ( $\Delta t$ ) of 0.025 are used. The boundary conditions for the problem considered are  $V_x = V_y = M = 0$  at  $S = 0$  and  $H = V = M = 0$  at  $S = 1$ . The cable shape (displacement response) is obtained by numerically integrating, by means of the central difference scheme, the velocity response computed.

Fig. 2 illustrates the cable shapes up to  $\bar{t} = 7$  at interval of 0.1, based on the box scheme. Note that near the free end, tension drops and the cable coils up into shape of high curvature. The modified box scheme gives similar results. Fig. 3 compares the cable shapes obtained by using the box scheme and modified box scheme. These results are in good agreement with Fried (1982) that presented numerical and experimental results for the forward motion phase and part of the backward motion phase.

## Cable Tension Response

Of importance to cable design is the cable tension, for example, in the study of fatigue of signal communication cables containing fiber optics. Somewhat surprisingly, the results of cable tension given by the box and modified box schemes are quite different. Specifically, high-frequency oscillation is found to exist in the dynamic tension response based on the box scheme but not on the modified box scheme. The comparison for cable tensions at the fixed end is given in Fig. 4. As Fried (1982) did not present cable tension, a physical experiment was therefore conducted using an actual cable (whose properties are different from the aforementioned problem).

## Experimental Setup

The cable used in the experiment is made of Neoprene rubber (homogeneous), with a solid circular cross section with a diameter of 25 mm and a density of  $1,430 \text{ kg/m}^3$ . By a simple tension test, Young's modulus  $E$  was found to be  $\sim 3.14 \times 10^6 \text{ N/m}^2$ .

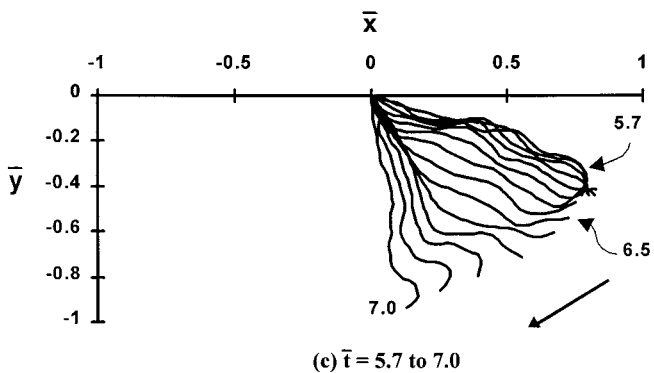
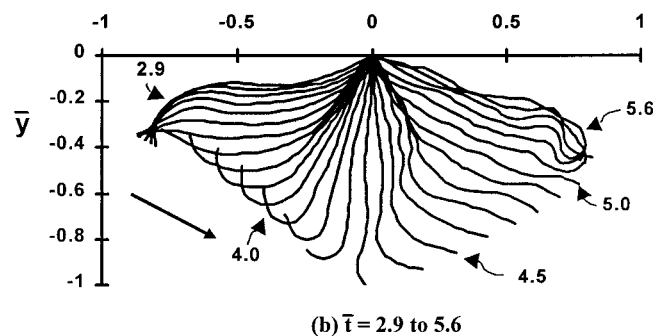
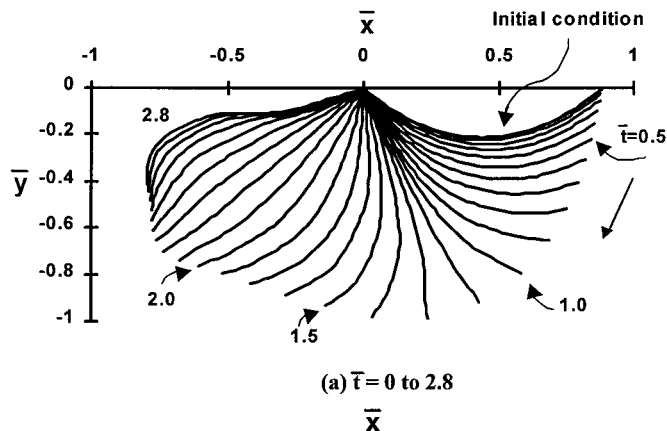


FIG. 2. Shapes of Cable Free Fall Based on Box Scheme

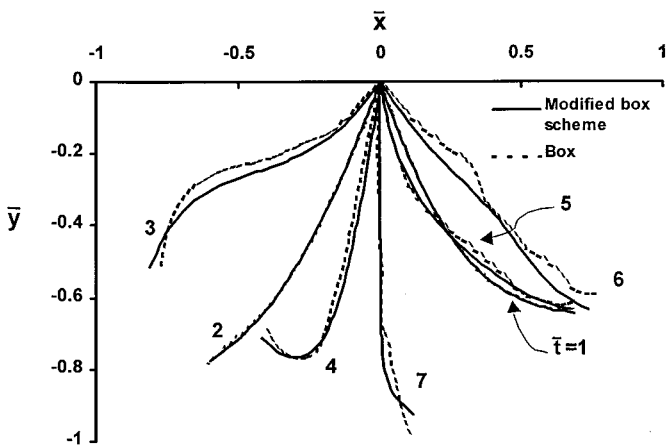


FIG. 3. Comparison of Cable Shapes Based on Box Scheme and Modified Box Scheme

Fig. 5(a) shows the experimental setup of cable suspended at two ends, each end having a small hole for insertion of a smooth pin. Between the two pins, the total cable length is  $L = 2.022$  m and the horizontal span is  $D_L = 1.805$  m. Near one (fixed) end, strain gauges are attached to provide a means of

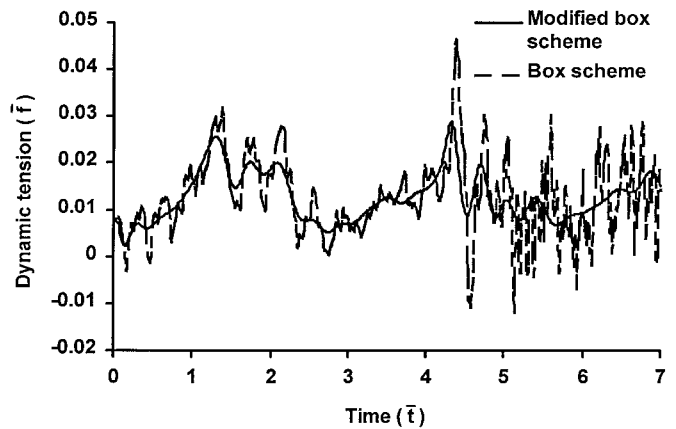


FIG. 4. Comparison of Dynamic Tensions Based on Box Scheme and Modified Box Scheme

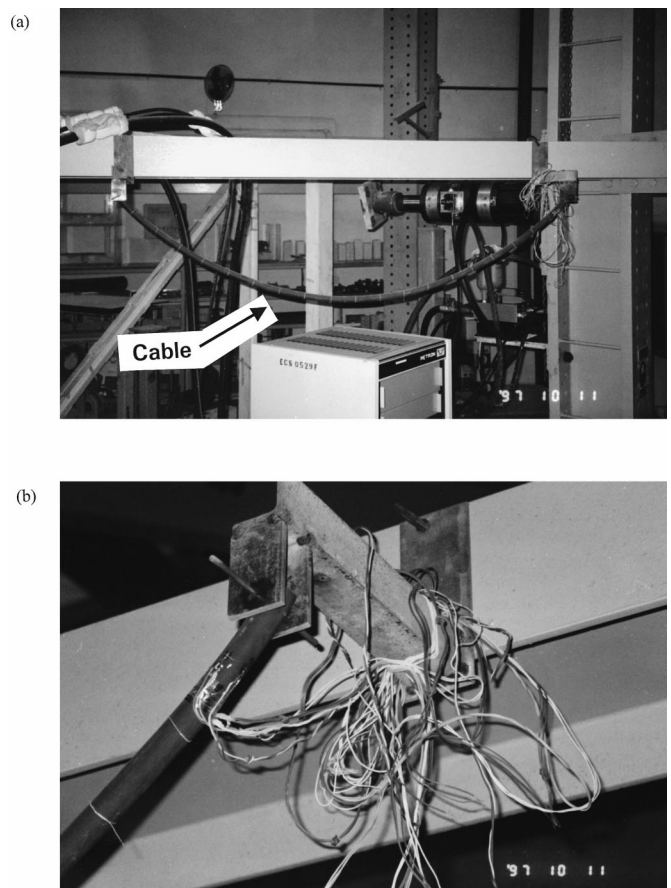


FIG. 5. Experimental Setup: (a) Cable Suspended at Two Ends; (b) Close Up of Strain Gauges near Fixed End

measuring tension at a location corresponding to the second node in the numerical model that has a total of 41 nodes. A close-up view of strain gauges attached at the fixed end of the cable is shown in Fig. 5(b). A total of four strain gauges are attached to the surface of the cable in a symmetrical manner as shown in Fig. 6. The strain gauges are located near the fixed end to minimize the movement effects of attached electrical cords but sufficiently far away from the pin hole to avoid the stress concentration zone.

To account for damping effect, a pull-back test was conducted by hanging a weight at the middle of the cable and suddenly removing it, thereby setting the cable into free vibration. From the amplitude decay of strains measured, the damping ratio is estimated to be  $\xi = 0.062$  (Zhang 1988). The

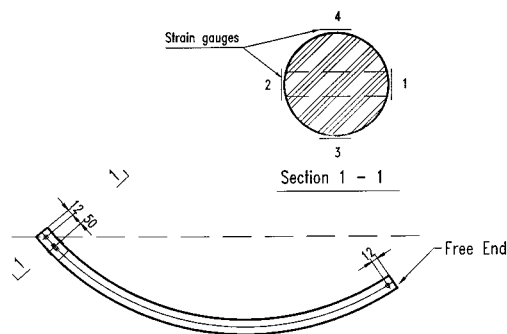


FIG. 6. Location of Strain Gauges

damping coefficient  $c$  is then obtained by using the following equation (Kahla 1995):

$$c = 2\xi A \sqrt{E\rho} \quad (38)$$

In the free-fall experiment, the pin at the “free” end is suddenly pulled off to allow this end to fall under gravity. The strain gauge readings are amplified by a dynamic strain meter, captured by a data acquisition system, and averaged out to give the tensile strain (for computation of cable tension). The experiment was repeated six times with various data acquisition rates of 1/256, 1/512, and 1/1,024 s. For each free-fall run, a total of 2,048 points were captured. It was noted that the experiment was highly repeatable, with different runs of free fall producing almost identical strain time histories.

For dynamic experiments, data acquisition rate plays a key role in the results. If the acquisition rate is not fast enough, the results may not truly reflect the dynamic response of the system. Strain response histories at acquisition rates of 1/256 and 1/1,024 s are shown in Fig. 7. It is seen that the strain response based on acquisition rate of 1,256 s is virtually the same as the result based on acquisition rate of 1/1,024 s. This shows that 1/256 s is fast enough to capture the dynamic tension response in the experiments.

For the box and modified box schemes, the numerical results for the dynamic tension response based on time step 1/200 s are found to be close to the results based on time steps shorter than 1/200 s. For the purpose of comparing the numerical results with the experimental results, the time steps used in the numerical procedures are taken to be the same as the experimental data acquisition rates (i.e., 1/256, 1/512, and 1/1,024 s).

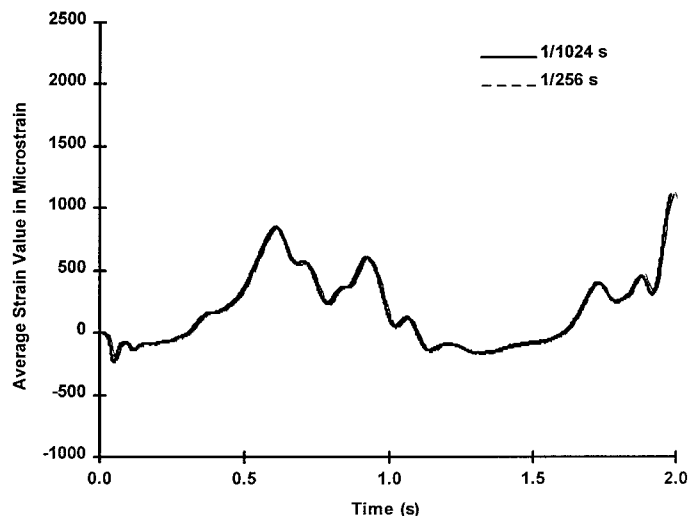


FIG. 7. Average Strain Values Recorded Using Two Different Data Acquisition Rates

### Comparison between Numerical and Experimental Results

The numerical results obtained by the box scheme using  $\Delta t = 1/256$  s are compared with the experimental results in Fig. 8. It is seen that the tension response contains high-frequency modes and oscillates around the test result before divergence at about  $t = 3.58$  s. The accuracy is adversely affected as a result of high-frequency oscillation. In contrast, the numerical results based on the modified box scheme using  $\Delta t = 1/256$  s agree well with the experimental results in general (Fig. 9).

To study the effects of time step on the numerical solutions, a smaller time step  $\Delta t = 1/1,024$  s is then used. The results are presented in Figs. 10 and 11. In these two figures, the experimental tension response was acquired at  $\Delta t = 1/1,024$  s. This acquisition rate is fast enough to capture the high-frequency mode obtained by the box scheme with  $\Delta t = 1/256$  s (Fig. 8), if it actually exists at all. With  $\Delta t = 1/1,024$  s, the numerical results given by the box scheme still contain high-frequency oscillation. As a matter of fact, it is noted that the box scheme with  $\Delta t = 1/1,024$  s gives tension response (Fig. 10) containing oscillation of frequency four times as high as the frequency of the high-frequency oscillation in the results based on  $\Delta t = 1/256$  s (Fig. 11). This confirms that the high-frequency oscillation as given by the box scheme is numerical

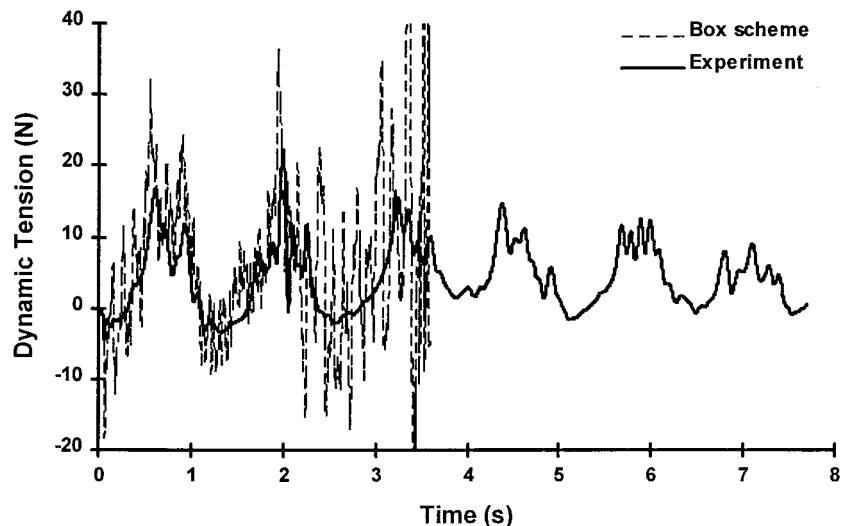


FIG. 8. Dynamic Tension Based on Box Scheme in Comparison with Experimental Results ( $\Delta t = 1/256$  s)

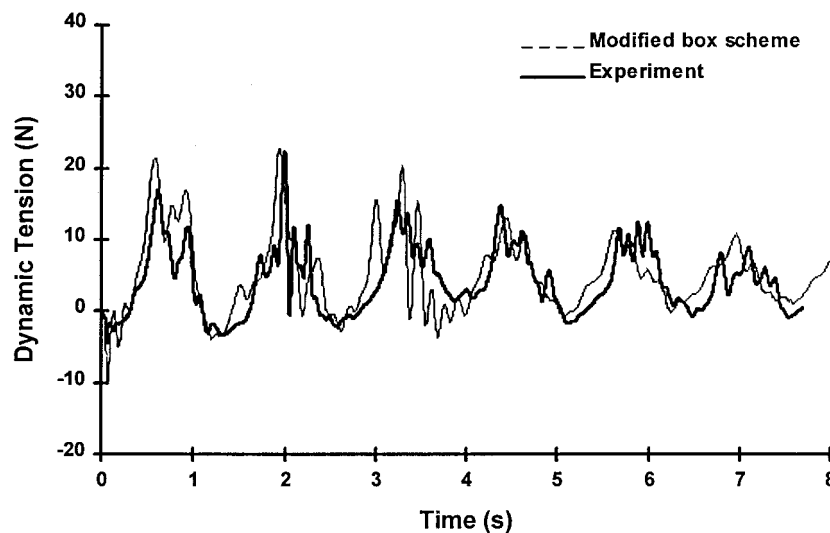


FIG. 9. Dynamic Tension Based on Modified Box Scheme in Comparison with Experimental Results ( $\Delta t = 1/256$  s)

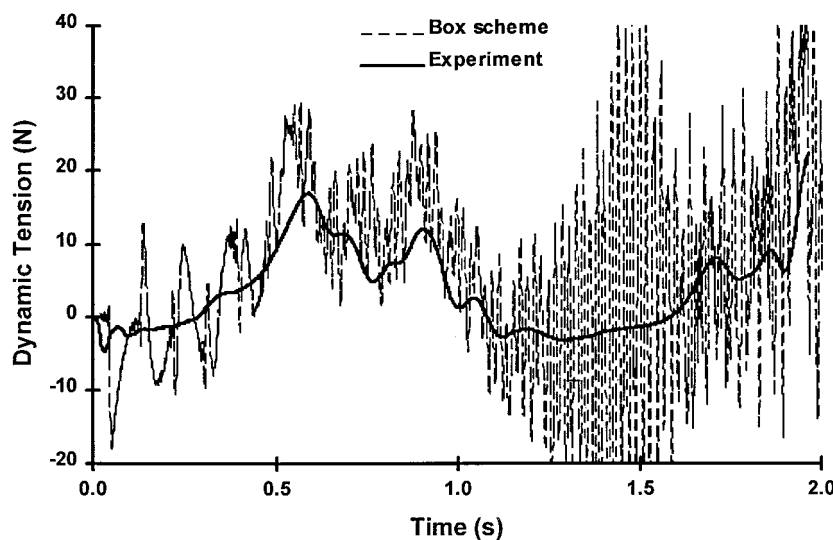


FIG. 10. Dynamic Tension Based on Box Scheme in Comparison with Experimental Results ( $\Delta t = 1/1,024$  s)

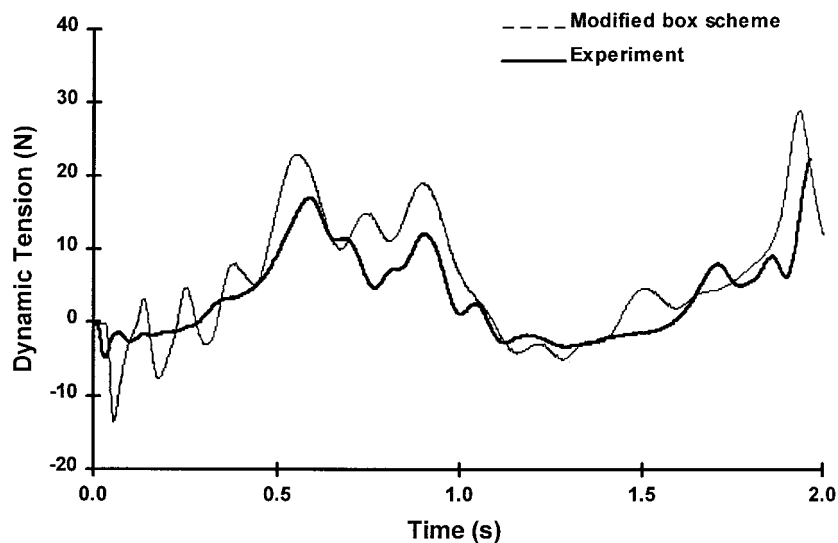


FIG. 11. Dynamic Tension Based on Modified Box Scheme in Comparison with Experimental Results ( $\Delta t = 1/1,024$  s)

rather than physical. The problem of the spurious high-frequency mode is eliminated with the use of the modified box scheme, which gives results in good agreement with the experimental results (Fig. 11).

### Efficiency and Convergence

Computational efficiency is an important consideration for numerical solution techniques. Because the box scheme con-

**TABLE 1. Computer Times (CPU s) per Time Step and Convergence**

| Time step<br>(1)    | $\Delta t = 1/256$ s<br>(2)                | $\Delta t = 1/512$ s<br>(3)                | $\Delta t = 1/1,024$ s<br>(4) |
|---------------------|--------------------------------------------|--------------------------------------------|-------------------------------|
| Time range computed | 8 s                                        | 4 s                                        | 2 s                           |
| Modified box scheme | 0.103 s/step (no divergence)               | 0.100 s/step (no divergence)               | 0.090 s/step (no divergence)  |
| Box scheme          | 0.172 s/step (divergence at $t = 3.582$ s) | 0.177 s/step (divergence at $t = 1.055$ s) | 0.157 s/step (no divergence)  |

tains high-frequency oscillation, it generally requires more iterations to achieve convergence and may even fail to converge after some time if the time step is not small enough. A Pentium 166-MHz personal computer is used to obtain the numerical results. The computer times [in central processing unit (CPU) seconds] per time step are summarized in Table 1. The proposed iterative scheme to solve the nonlinear equations has been used for both the box scheme and modified box scheme in this comparison. The modified box scheme is roughly 60% faster than the box scheme.

With regard to solving nonlinear equations, the proposed iterative method markedly improves the efficiency of the method. For comparison purpose, IMSL (1989) software (Visual Numerics, Inc., Houston, Tex.), which is widely used in science and engineering, is also used here to solve the simultaneous nonlinear equations [i.e., (18)]. An IMSL subroutine called DNEQNF is employed, which implements a variation of Newton-Raphson's method and uses FD approximation for the Jacobian matrix.

For this particular problem, the proposed iterative procedure is found to be faster than the IMSL subroutine to achieve roughly the same accuracy. On a DEC 4000/820 workstation (160 MHz), the modified box scheme with the iterative method takes about 51 s of CPU time to obtain 8 s of dynamic response at  $\Delta t = 1/256$  s, whereas the use of Newton-Raphson's method by an IMSL subroutine DNEQNF takes about 1,430 s ( $\sim 28$  times longer). It should be noted that the box scheme fails to converge (at  $t \approx 3.58$  s) irrespective of whether the iterative method or Newton-Raphson's method is used.

## CONCLUSIONS

This paper deals with the modeling of the large displacement motion of low-tension cables. The equations of motions are derived in a global coordinates system, accounting for axial and flexural deformations. An efficient and robust scheme based on the FD method, namely, the modified box scheme with an iterative procedure for solving the nonlinear equations, is developed to solve the problem numerically. This scheme is a modification of an existing FD procedure known as the box scheme. A large displacement motion problem of cable free fall is investigated. Both the box scheme and modified box scheme give approximately the same results for cable shapes that are in good agreement with published results. The two schemes, however, give different results of dynamic tension response. To verify the accuracy and, in particular, the existence (or otherwise) of high-frequency oscillations in cable tension, a physical experiment was conducted. By comparing the box scheme's results using different time steps and also the experimental results using different acquisition rates, it is concluded that the high-frequency mode dependent on the time step used is a numerical and not a physical phenomenon. The modified box scheme eliminates the spurious high-frequency mode and gives cable tension response in good agreement with

the experimental results. In terms of numerical convergence and speed, it is also superior to the box scheme. The proposed iterative method for solving nonlinear equations renders the modified box scheme efficient, and it is faster than the use of the commonly used Newton-Raphson's method. The method has also been used successfully to solve some other cable dynamics problems (Zhang 1998).

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